In class lab #10 for Numerical Analysis

Your name: _____

Submit both your code and results before 23:59 Thur, May 16 or you will get 0.

Runge-Kutta Method

The method we will study is Runge-Kutta Method for an initial value problem for a first-order ordinary differential equation,

$$\begin{cases} y' = f(t, y) \\ y(a) = y_a \\ t \text{ in } [a, b]. \end{cases} \tag{1}$$

We begin with a grid of $n + 1$ points

$$t_0 < t_1 < t_2 < \cdots < t_n$$

along with the t-axis with equal step size h.

Runge-Kutta Method:

$$\begin{cases} y_{n+1} = y_n + h \sum_{i=1}^{p} c_i K_i \\ K_1 = f(t_n, y_n) \\ K_i = f(t_n + a_i h, y_n + h \sum_{j=1}^{i-1} b_{ij} K_j)(i = 2, 3, \cdots, p) \end{cases} \tag{2}$$

a_i, b_{ij}, c_i are parameters.

Then we give the Runge-Kutta Method of order four(RK4):

$$\begin{cases} y_{n+1} = y_n + \frac{h}{6}(K_1 + 2K_2 + 2K_3 + K_4) \\ K_1 = f(t_n, y_n) \\ K_2 = f(t_n + \frac{h}{2}, y_n + \frac{h}{2} K_1) \\ K_3 = f(t_n + \frac{h}{2}, y_n + \frac{h}{2} K_2) \\ K_4 = f(t_n + h, y_n + h K_3) \end{cases} \tag{3}$$

Problem 1. (50 points) Consider the initial value problem of the following differential equation

$$\begin{cases} y' = 2(t + 1)y \\ y(0) = 1. \end{cases} \tag{4}$$

- Apply the fourth-order Runge-Kutta Method solution on a grid of step size $h = 0.1$ in $[0, 1]$ for the initial value problem. Print a table of the t value, approximations, and global truncation error at each step.
  (global truncation error:$g_i = |y_i - y(t_i)|, y(t_i)$ is the correct solution)

Problem 2. (50 points) Plot the fourth-order Rungr-Kutta Method approximate solution on [0,1] for the differential equation $y' = 1 + y^2$ and initial condition (a) $y_0 = 0$ (b) $y_0 = 1$, along with the exact solution (a) $y = \tan(t)$, (b) $y = \tan(t + \pi/4)$. Use step sizes $h = 0.1$ and $h = 0.05$.